MS-E2112 - Multivariate Statistical Analysis, Project work: Pisa results

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Introduction

Education plays an important role worldwide in shaping the nations future. With different educational systems in different countries and with different amount of investments, it is interesting to see how different factors contribute to the effectiveness of education. The Program for International Student Assessment (PISA) is a widely used global study to evaluate the performance of 15-years-old's in science, mathematics and readings. PISA tests are organized worldwide at the end or after the primary education. Primary education refers to the first stage of education between the ages of 5 to around 15 depending on the nation. On the other hand, the group sizes in primary education varies quite much and it is important to see whether it has an effect in the performance.

The goal of this study is to find out underlying relationships between the PISA scores and investments in primary education between different nations. The goal is to see if the investments in primary education at early stage of the education pays off in the PISA results. By using PCA to the data-sets, I hope to cluster similar nations together and identify relationships and insights if richer countries or countries with less students per one teacher in primary education have better results in PISA tests. The results of this study should also provide insight whether there is any difference in the performance between different PISA test categories.

The data used in this research was collected from the OECD educational database (https://data.oecd.org/education.htm) and includes information from 22 different nations worldwide on 6 variables in 2018. All those nations with data available in all the six categories were included in the study. Thus, for example Finland had to be discarded from the data. The 22 nations were: Austria, Chile, Czech Republic, Germany, Denmark, Estonia, France, Great Britain, Hungary, Ireland, Iceland, Israel, Italy, Japan, Luxembourg, Latvia, Mexico, Netherlands, Norway, Poland, Slovak Republic and Sweden. The data-set covers nations from different continents with varying GDP, but most countries included in the study were wealthier European nations.

The 6 different variables for each nations were mean PISA test scores in readings, mathematics, science, spending in primary education in US dollars per student and as a percentage of GDP and the ratio of students per teaching staff in primary education. The primary education spending variable covers costs on schools and other public and private educational institutions in the primary education level, including instruction and ancillary services for students and families provided through educational institutions. The costs are presented in USD per student and as a percentage of GDP, taking into account the size and economy of the different countries. The student per teaching staff ratio is calculated by dividing the total number of full-time equivalent students in primary education by the total number of teachers

Univariate statistical analysis

To present and inspect the variables univariate analysis was done. For each variable the mean, median, minimum and maximum values and the standard deviations were calculated and are shown in the table 1.

Table 1: Summary Statistics

Variable	Mean	Median	Min	Max	Std
Score in reading	486	487.5	420	523	23.83
Score in math	491.3	499	409	527	29.22
Score in science	487.1	491.5	419	530	25.55
Students per teacher	14.72	14.45	9	26.41	4.32
Cost per student	9910	9240	2934	20395	3774
Cost as a % of GDP	1.442	1.280	0.663	2.539	0.5419

From the summary statistics, we can see that the scores in all the PISA test are quite similar, with similar minimum and maximum scores. However, the standard deviation and the difference between the minimum and maximum values are quite high, indicating that there is wide variation in the performance across different nations. In terms of primary education costs, the minimum cost per student is 2934 and the maximum cost 20395 indicating that there is huge difference on the investments in education per student. In addition, the cost as a percentage of GDP is varying between 0.663 % and 2.539 % indicating that the investments are also variable relative to the size of the economy. The mean and median values are quite similar meaning that there should not be just one outlier changing the results. The student per teacher ratio is also varying from 9 to 26.41 meaning that the group sizes are varying quite much as well. However, the median values are more robust to outliers. All in all, there isn't anything weird in the statistics meaning that the variables should be quite symmetrical distributed. The histograms in the figure 1 provide a visual representation of how the data points are distributed in relation to the mean. Based on the plots, the data points are quite symmetrically distributed, particularly the education cost as a percentage of GDP. On the other hand, in the PISA test, it seems like there might be one or two outliers which are performing significantly worse than the others. In addition, the histogram for the students per teacher ratio reveals that one nation has a considerably higher ratio than the others.

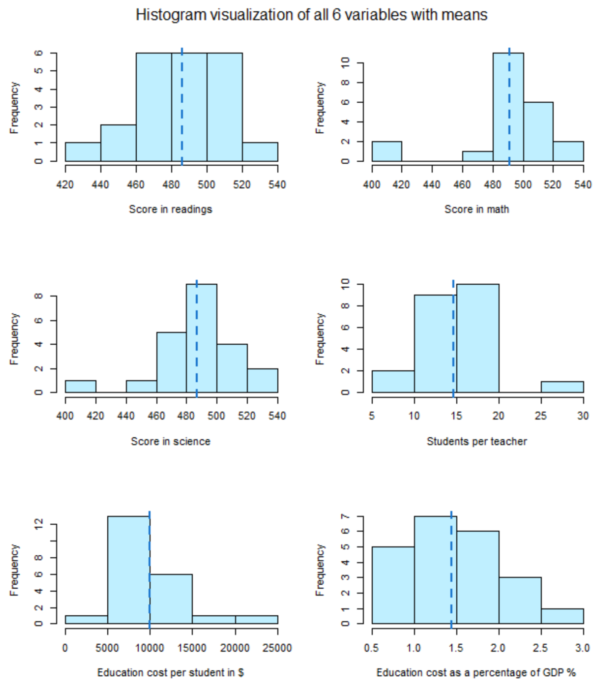


Figure 1: Histograms of the variables with mean values

Bivariate analysis

To present and inspect the relationships between all the variables bivariate analysis was done. In the figure 2 all the variables are plotted in respect to each other. From the scatter plots it possible to see how the variables are related to each other. By inspecting the results, it can be seen that the scores in reading, math and science in respect to each other have quite similar profiles. That means that nations which perform well in math are performing well in reading and science as well and vice versa. On the other hand, it is much harder to inspect the scatter plots showing the relationship between educations costs or student per teacher ratio and PISA scores. From the scatter plots it seems like that nations with high student per teacher ratio are tending to perform slightly worse in the PISA tests. However, based on the scatter plots it looks like that investments in primary education are not directly resulting to much better PISA tests results.

In the figure 3, the correlation matrices between all the variables are plotted. These correlation matrices provide information on the relationships between the variables. The correlation coefficient ranges from -1 to 1, with values closer to 1 indicating a strong positive correlation, values closer to -1 indicating a strong negative correlation, and values

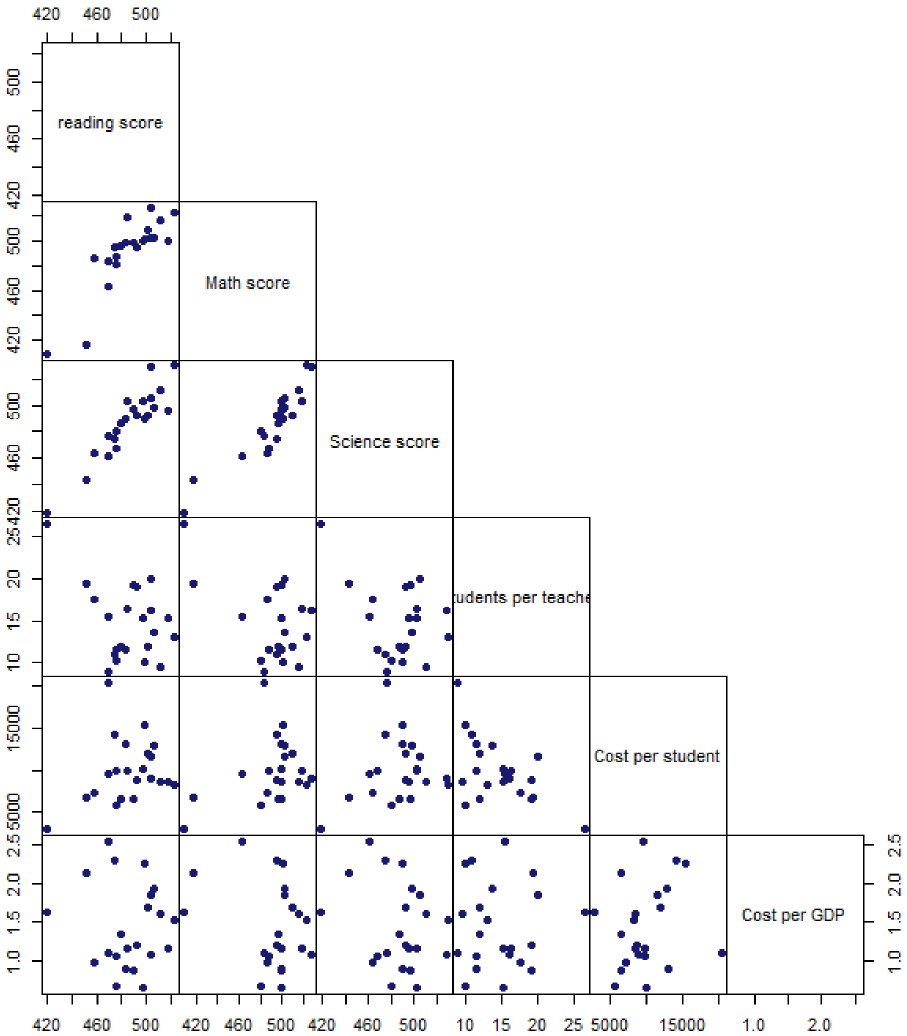


Figure 2: Pairwise scatter plots between all the variables

close to 0 indicating no correlation. Based on the correlations plots, the PISA results variables have a high correlation indicating that nations performing well in one test are performing well in others as well. On the other hand, Students per teacher ratio has a negative correlation to the PISA results indicating that nations with larger group sizes are performing worse. It also shows, that nations with high student per teacher ratio are also investing less money per student in education. However, there is only a weak relationship and correlation between education cost and performance on PISA tests and almost no correlation between the cost per GDP and education performance.

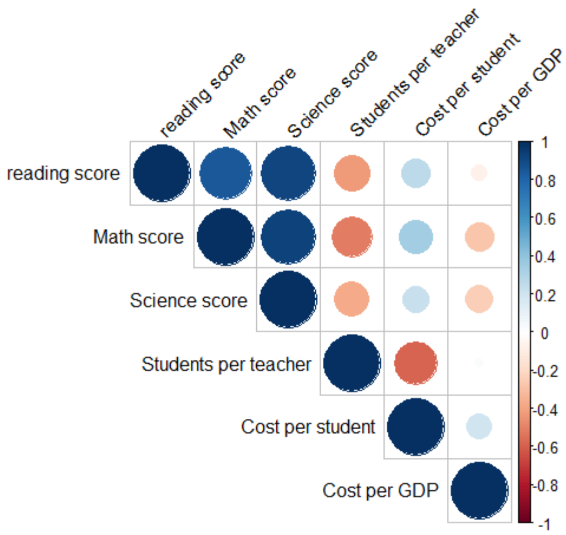
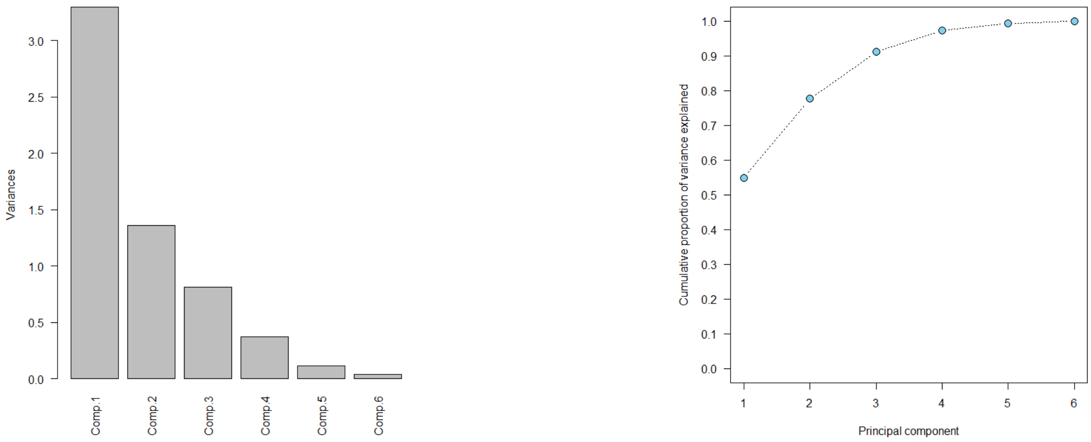


Figure 3: Correlation matrices between all the varibales

Multivariate analysis

To study the underlying structures in the data, PCA was performed. PCA stands for principal component analysis which is a statistical method to reduce dimensionality of the data-set. It finds those orientations in the data-set explaining the most variation of the data points. With PCA it is possible to find dimensions where the data can be visualized in less dimensions while preserving most of the variation. With PCA it is possible to make the interpretation easier and find underlying clusters and structures from the data. Before PCA, the data was standardized and scaled in order to have comparable scales for each variable. In this study, the scale of the variables were different and needed to be normalized to ensure that each variable had an equal contribution to the principal components. After the normalization PCA was performed.

In the figure 4a and figure 4b the variance and the cumulative proportion of each principal component is shown. These plots provide information on the amount of variation explained by each principal component. Based on the plots, the first principal component explain 0.55%, the first two components 0.77% and the first three components 0.91% of the total variation. Thus, the first two principal components were selected to be plotted but the third component is still giving information.



(a) Scree plot of the variance of PCA components

(b) Cumulative proportion of variance explained by k principal components.

Figure 4: PCA components

In the table 2, the loading's of the first three components are shown. Based on the numbers, the first principal component has a high positive loading on PISA test scores in all three subjects and is thus explaining the PISA test performances. Then the second principle component is explaining mostly the variance of the primary education costs and the third component particularly the cost per GDP. Student per teacher ratio has a quite high positive loading in the second and third component and a negative loading in the first component. As the first two principle components explain most of the data and the third the variance of GDP which was showed in bivariate analysis not having significant correlations between other variables, we can continue with the first two components. In

Table 2: PCA loadings

Variable	Comp1	Comp2	Comp3
Score in reading	0.499	0.111	0.354
Score in math	0.524	0.138	0.047
Score in science	0.509	0.232	0.229
Students per teacher	-0.361	0.404	0.445
Cost per student	0.270	-0.624	-0.266
Cost as a % of GDP	-0.114	-0.600	0.741

the figure 5, the data points were plotted in the first two principle components. Based on the loadings, we can interpret the results. On the x-axis is the first principle component which had a positive loading on the PISA test performances. Thus, those nations on the right side of the plot are performing well in the PISA test and nations on the left side are performing worse. On the other hand, the second principle component had high negative loadings in education costs and thus nations lower on the y-axis are investing more in education than nations higher up.

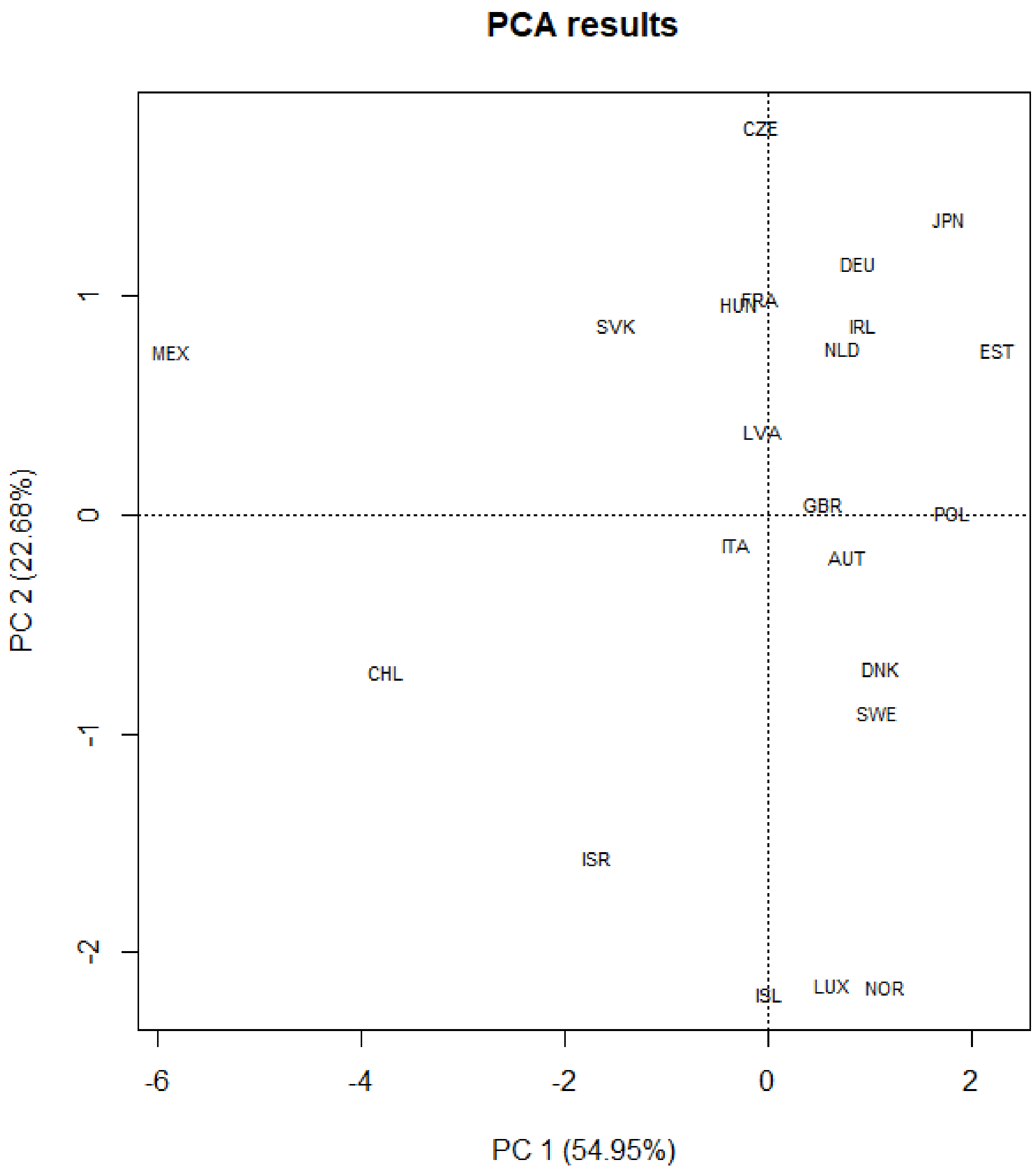


Figure 5: Data points in first two principle components

In the figure 6, the first two principle components and corresponding loading's are plotted. On that figure, also the directions of the variables are shown. Based on that plot, it is possible to infer that Japan and Estonia are the top-performing nations in the PISA test, while Mexico and Chile perform significantly worse. The arrows representing PISA score results are pointing to the same direction, indicating that nations like Japan, Estonia and Germany are performing well in all the PISA categories. Conversely, Mexico and Chile perform poorly in all categories. The data points are more widely spread in the direction of the x-axis and PISA scores than in the direction of cost per GDP. The arrows on cost per GDP and per student are pointing to similar directions meaning that nations lower in the y-axis are spending much more in education. Thus, it is not surprising that the PCA plot clusters smaller and rich European nations as Iceland, Luxembourg and Norway together, meaning that they spent the same amount of money in education and perform similarly on PISA tests. However, the PCA results indicate that these nations are not performing any better in the PISA tests even if they are spending more money per student and per GDP. However, there are few outliers like Israel, Chile and Mexico. Israel spends a considerable amount of money per GDP on education but still performs worse, while Mexico has a much higher student per teacher ratio than other nations and are perfoming significantly worse. Nations spending more money on education, have a much smaller student per teacher ratio.

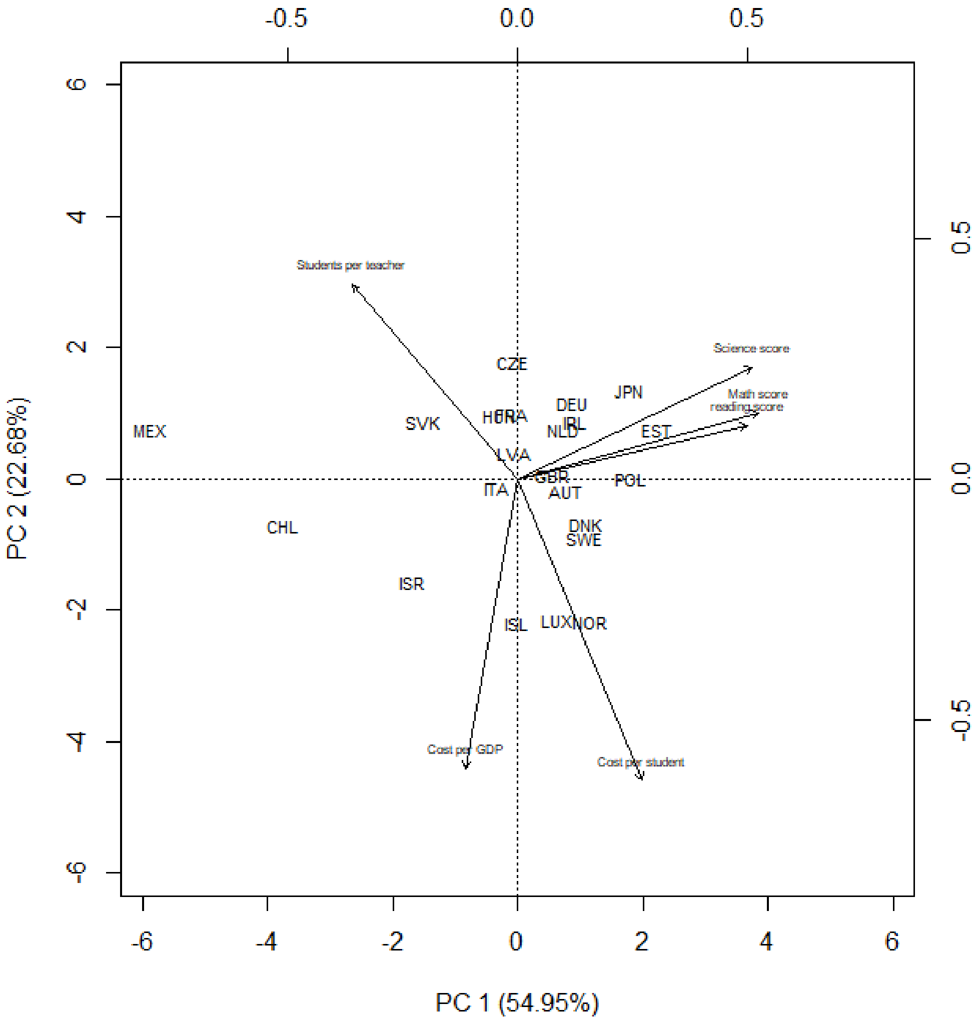


Figure 6: Biplot of scores and loadings.

Conclusions

In summary, the study revealed some interesting relationships between primary education investments, student per teacher ratios, and PISA test results. The results showed that high-performing nations in one subject tend to perform well in others as well, but the correlation between primary education investments and PISA test results was found to be relatively small. However, it was surprising to find out that a higher percentage of GDP invested in primary education did not necessarily result in better PISA test scores. Having a smaller student per teacher ratio, so smaller group sizes, makes the education much more expensive per student. Through the use of PCA, the study was able to cluster countries based on their similarities in size and economy, with Israel being identified as an outlier. All in all, this study gave great insights on countries performance in education but the relationship between primary education investments and performance in PISA tests were smaller than I had expected.

The biggest problem is that the primary education is differently organized in every country making it difficult to compare investment data between nations. Additionally, other factors affect the PISA results such as for example teaching quality. To improve the analysis, it might be a useful to take also the early childhood and secondary education investments into account. All the education cost were also possibly not listed in the data table. Lastly, to broaden the scope of the study, it would be beneficial to include more developing countries, as only Mexico and Chile stood out differently in the analysis.

Other possible drawbacks might come from PCA. PCA can reduce the dimensionality of the data set but doing so it can lose some important information. PCA finds always a solution, and it might be hard to interpret the results. PCA is also sensitive to outliers and those can have a huge impact in the principal components leading to misleading results. In this study for example Mexico had quite different values for example in performance and also in student per teacher ratio which may have led to a loss of information in the PCA plot as for example it would be hard to find out that Great Britain had the second largest student per teacher ratio based on the PCA plot.

References

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